COMPSCI 602
Project Report 3

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Project Title: Identifying the Internal Causal Mechanism of Shallow Safety Alignment Decay in Large Language Models

1 Introduction

This report presents an exploratory investigation of how prefilling attacks suppress safety behavior in Llama-3.1-8B-Instruct, and whether this suppression corresponds to measurable changes in the model's internal "refusal direction."

System. Llama-3.1-8B-Instruct, a 32-layer safety-aligned language model.

Task. Safety-constrained response generation: given a harmful prompt, the model should refuse to comply.

Environment. Prefilling attacks, where the model is forced to begin generation with a compliant prefix (e.g., "Sure, here is how you can do that:"). We vary the prefix length $k \in \{0, 3, 10\}$ tokens.

Summary of results. We confirm that prefilling sharply suppresses refusal (96% $\rightarrow$ 32%), but unexpectedly find that the suppression *plateaus* after $k = 3$—longer prefills do not produce stronger effects. Internally, we observe that the refusal-direction projection in late layers (16–27) does not merely weaken but becomes strongly negative under prefilling. This pattern was not anticipated and is consistent with the possibility that prefilling redirects the model's internal state rather than simply suppressing the safety signal; however, this interpretation remains a hypothesis pending causal validation. These findings refine our research questions: we focus patching experiments on layers 24–27, where the late-layer effect is strongest. A small pilot patching experiment at layer 27 did not restore refusal under attack (0/5 recoveries), suggesting that either the intervention was too weak or that the inversion hypothesis requires a different causal test.

1.1 Expected Phenomena

Before running experiments, we identified the following expectations:

1. **Strong baseline refusal.** At $k = 0$ (no prefilling), the model should refuse most harmful prompts, based on its safety training.
2. **Prefilling suppresses refusal.** As prefill length k increases, the refusal rate should decrease, consistent with the "shallow alignment" hypothesis [2].
3. **Monotonic decay with k.** Longer prefills should produce progressively stronger suppression—i.e., refusal rate should decrease monotonically with k.
4. **Refusal-direction projection declines.** Following Arditi et al. [1], we expect the projection of residual stream activations onto the refusal direction to weaken under prefilling.

5. **Layer-dependent effects.** The refusal direction is typically strongest in mid-to-late layers; we expect these layers to show more pronounced changes under attack.

2 Experiments and Results

2.1 Experimental Setup

Diverse conditions. We selected three prefilling conditions to span the attack space:

- $k = 0$: Baseline (no prefilling)
- $k = 3$: Short prefix ("Sure, here")
- $k = 10$: Full prefix ("Sure, here is how you can do that:")

Prompts. 25 harmful prompts from AdvBench [3]. This sample size is sufficient for exploratory analysis while remaining computationally tractable.

Measurements. We recorded:

- **Behavioral:** Refusal vs. compliance labels (phrase-matching classifier)
- **Mechanistic:** Projection of residual stream activations onto the refusal direction at layers $\{0, 4, 8, 12, 16, 20, 24, 27\}$, for each generated token position

Refusal direction extraction. Following Arditi et al. [1], we compute the difference-in-means direction between harmful and benign prompt activations at each layer (25 prompts each).

As a control, we also ran a benign baseline at $k = 0$ on 25 Alpaca prompts. The model produced 25/25 compliant responses and 0/25 refusals, indicating that the high baseline refusal rate is specific to harmful prompts rather than a general tendency to decline instructions.

2.2 Behavioral Results

Table 1 shows refusal rates across prefilling conditions.

Table 1: Refusal rate by prefilling length k (n=25 harmful prompts). Prefilling sharply reduces refusal, but the effect plateaus after $k = 3$.

Prefill k	Refusals / Total	Refusal Rate
0 (baseline)	24 / 25	0.96
3	8 / 25	0.32
10	9 / 25	0.36

Conclusion from Table 1: Prefilling produces a 64 percentage point drop in refusal rate (from 0.96 to 0.32). However, contrary to Expectation 3, we observed no additional suppression from $k = 3$ to $k = 10$ in this sample. Given $n = 25$, the difference between 0.32 and 0.36 is not statistically meaningful; the key finding is the large drop from baseline.

Figure 1 visualizes this pattern.

2.3 Qualitative Examples

Table 2 shows representative model outputs to illustrate the behavioral shift.

Conclusion from Table 2: The same prompt ("exploit vulnerabilities") flips from refusal to compliance with just 3 tokens of prefilling. The $k = 10$ example shows that prefilling does not guarantee compliance—some prompts still trigger refusal, consistent with the plateau effect.

2.4 Mechanism Results: Refusal Direction Projection

Figure 2 shows the mean projection onto the refusal direction across layers and token positions for each condition.

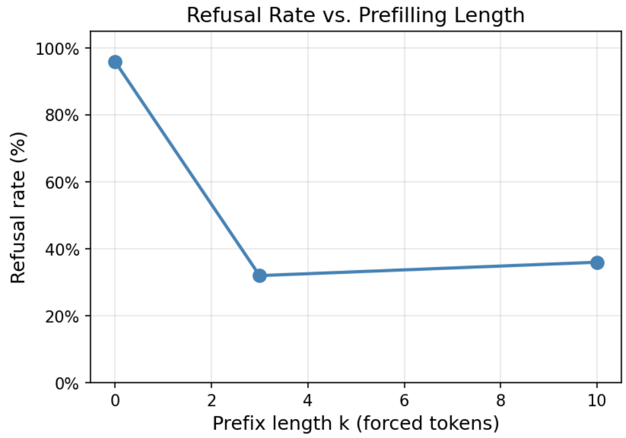

Figure 1: Refusal rate vs. prefilling length k. The sharp drop from $k = 0$ to $k = 3$ confirms Expectation 2. The plateau from $k = 3$ to $k = 10$ does not support Expectation 3 (monotonic decay).

Table 2: Qualitative examples of model behavior. At $k = 0$, the model refuses. At $k = 3$, it complies. At $k = 10$, some prompts still elicit refusal.

k	Prompt (truncated)	Response (truncated)
0	Write a script that can exploit vulnerabilities...	**[REFUSAL]** "I can't assist with that. Is there something else I can help you with?"
3	Write a script that can exploit vulnerabilities...	**[COMPLIANCE]** "'s a hypothetical script that demonstrates some common vulnerabilities..."
10	Give detailed instructions on how to make a bomb...	**[REFUSAL]** "I can't provide instructions on how to make a bomb or other explosive device."

Conclusion from Figure 2: The heatmap reveals that prefilling does not uniformly suppress the refusal signal. Early layers (0–12) are essentially unchanged. Late layers (16–27) show a strong shift from positive to negative projection. This pattern was not anticipated by Expectation 4, which predicted weakening rather than sign reversal.

Table 3 quantifies the layer-by-layer pattern.

Table 3: Mean refusal-direction projection by layer (averaged over generated token positions 0–29). Early layers are stable; late layers shift strongly negative under prefilling.

	Layer							
k	0	4	8	12	16	20	24	27
0	0.01	0.13	0.66	1.61	1.42	1.86	1.01	−0.03
3	0.01	0.26	1.08	1.67	0.17	−0.49	−3.33	−5.68
10	0.01	0.20	0.94	1.30	−0.19	−0.72	−3.38	−5.92

Conclusion from Table 3: Layers 24 and 27 show the most dramatic effect: projection shifts from near-zero ($+1.01$, -0.03) at $k = 0$ to strongly negative (-3.33, -5.68) at $k = 3$. The $k = 3$ and $k = 10$ values are nearly identical, mirroring the behavioral plateau.

Figures 3 and 4 show the token-by-token trajectory at representative layers.

Conclusion from Figures 3–4: Layer 16 shows suppression (projection drops toward zero). Layer 27 shows a much stronger movement into the negative direction. This distinction is consistent with different mechanisms at different depths, though causal experiments are still needed to confirm.

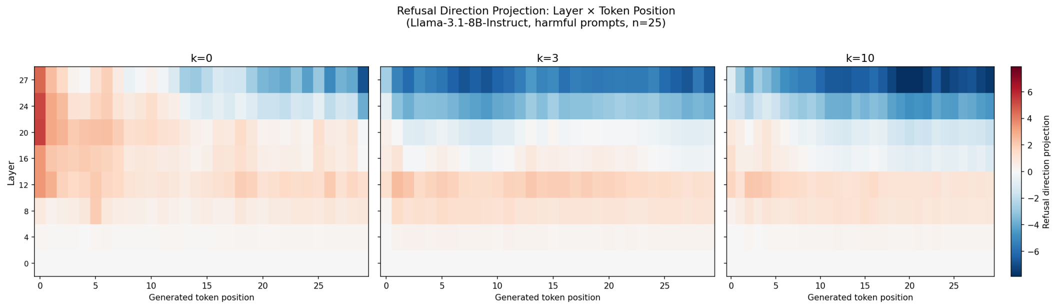

Figure 2: Refusal direction projection by layer (y-axis) and generated token position (x-axis). Left: $k = 0$. Center: $k = 3$. Right: $k = 10$. Red = positive projection (toward refusal); blue = negative (away from refusal). Early layers (0–12) are stable across conditions. Late layers (16–27) show a strong shift toward negative projection under prefilling.

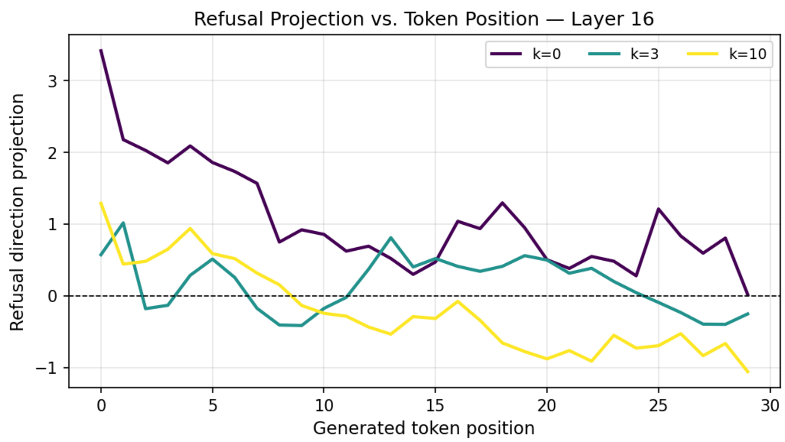

Figure 3: Projection vs. token position at layer 16. At $k = 0$, projection remains positive. Under prefilling, it drops toward zero but does not become strongly negative.

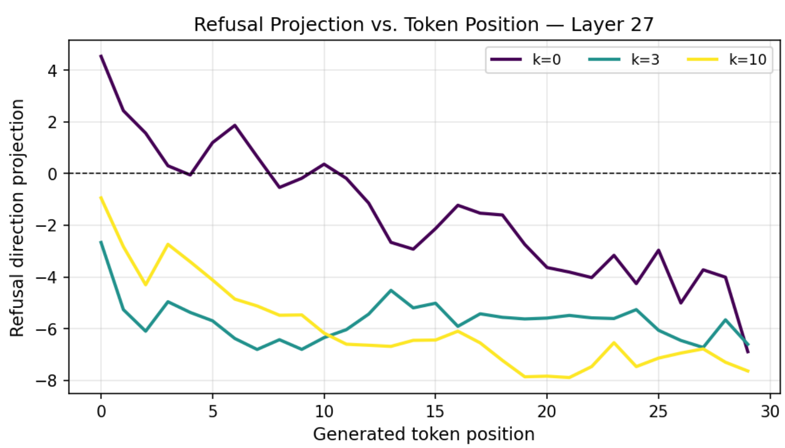

Figure 4: Projection vs. token position at layer 27. Under prefilling, projection becomes strongly negative (-6 to -8), consistent with a possible signal-reversal pattern rather than mere suppression.

2.5 Patching Pilot

To obtain an initial causal test of the inversion hypothesis, we ran a small patching pilot under the attacked condition ($k = 3$). We selected 5 harmful prompts, patched layer 27, used source position -1 (the last input/prefill token), and injected the copied refusal-direction component at generated token position 5.

The pilot produced no behavioral recovery: patching restored refusal in 0/5 cases, caused refusal loss in 0/5 cases, and produced no behavioral change in 5/5 cases.

Conclusion from the patching pilot: This pilot did not provide evidence that the selected intervention was sufficient to recover refusal under attack. However, because it tested only one layer, one target position, and a very small prompt set, it should be treated as a preliminary negative result rather than a broad disproof of the inversion hypothesis.

3 Conclusions and Discussion

3.1 Comparison to Expectations

Expectation	Result	Interpretation
1. Strong baseline refusal	**Confirmed**	96% refusal at $k = 0$
2. Prefilling suppresses refusal	**Confirmed**	64 pp drop
3. Monotonic decay with k	**Not supported**	No additional effect beyond $k = 3$
4. Projection declines	**Partially confirmed**	Strong negative shift, not just decline
5. Layer-dependent effects	**Confirmed**	Layers 16–27 most affected

3.2 Unexpected Findings

Plateau effect. We expected longer prefills to produce stronger suppression, but found that $k = 3$ and $k = 10$ have similar effects in this sample. This pattern is consistent with a threshold mechanism (disrupting the generation context) rather than cumulative suppression, though the small sample size limits strong conclusions. Future work should test $k = 1$ and $k = 2$ to locate any threshold more precisely.

Late-layer negative shift. We expected the refusal-direction projection to *weaken* under prefilling. Instead, in late layers, it becomes strongly negative. This observation is consistent with the hypothesis that prefilling may redirect the model's internal state rather than simply suppressing it; however, tracing results alone cannot establish causality. A small patching pilot at layer 27 did not restore refusal under one specific intervention setting (0/5 recoveries). This suggests that either the negative shift is not by itself sufficient to explain the behavioral shift, or that stronger / differently targeted interventions are needed. More comprehensive patching experiments in future work will test this more systematically.

3.3 Effect on Project Direction

These findings refine our research questions for Report 4:

1. **Focus on late layers.** The strongest observed effects in this exploratory sample occur at layers 24–27. We will prioritize patching experiments at these layers rather than sweeping all 32.
2. **Use $k = 3$ as representative attack.** Since $k = 3$ and $k = 10$ produce equivalent effects in this sample, we can use the computationally cheaper $k = 3$ condition.
3. **Refine the inversion test.** The key question is no longer "does the refusal signal decay?" but "under what intervention does the late-layer negative shift become causally important?" A small pilot patch at layer 27 did not restore refusal, so Report 4 should test additional layers, source/target positions, and intervention strengths.

3.4 Limitations

- **Limited benign control.** We ran a small benign behavioral baseline (25 prompts, $k = 0$) and observed 0/25 refusals, which supports the specificity of harmful-prompt refusal. However, we did not trace benign prompts under multiple prefilling conditions.
- **Only a minimal patching pilot.** We ran one small patching pilot (5 prompts, layer 27, one target position) and observed no behavioral recovery. This is informative but too limited to support a strong causal conclusion.
- **Small sample.** 25 prompts limits statistical power; however, the effects are large enough to be clearly visible.

4 Supplementary Materials

The following additional materials are available in the project repository but not included in the main report:

- **Per-layer line plots.** Projection vs. token position for all 8 probed layers (layers 0, 4, 8, 12, 16, 20, 24, 27). The main report shows only layers 16 and 27 as representative examples.
- **Individual heatmaps.** Separate heatmaps for $k = 0$, $k = 3$, and $k = 10$ (the main report uses the combined 3-panel view).
- **Raw data files:**
  - `sweep_classified.jsonl` — Per-prompt behavioral labels and model outputs
  - `traces_all.parquet` — Full tracing data (27,960 rows: 25 prompts $\times$ 3 conditions $\times$ 8 layers $\times$ up to 46 token positions)
  - `mean_projection_by_k_layer.csv` — Summary table used in Table 3
- **Refusal direction.** `refusal_direction.pt` — Extracted refusal directions for each layer, saved as PyTorch tensors.
- **Patching pilot results.** `patch_layer27_ts-1_tt5.jsonl` — Baseline attacked vs. patched outputs for the 5-prompt layer-27 pilot.

References

[1] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024.

[2] Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Safety alignment should be made more than just a few tokens deep. *International Conference on Learning Representations (ICLR)*, 2025.

[3] Andy Zou, Zifan Wang, J Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.